

1 Polynomials - more exercises

1. Let $p(x) = 3x^4 + 4x^3 + 6(n-2)x^2 - 12nx + n^2$.
 - (a) Find values of n s.t there is only one stationary point of p .
 - (b) If p has only one stationary point, find values of n for which $p(x) = 0$ has no solutions
2. Find values of p such that the function $f(x) = px^2 + 13x + p$ and $y = 12x + 3p$ only has one intersection
3. (CAS) State the set of values of b s.t $2x^3 + 3x^2 - 5x + b = 0$ has 3 solutions
4. Let $f'(x) = 2ax^3 + 3bx^2 - 6ax + b$ such that f is convex on the interval $-1 < x < 1$ and concave elsewhere, and $f''(0) = 1$. Find possible values of a, b
5. Let $f(x) = 3x^3 + 2x^2 + x$
 - (a) Find the smallest angle between the tangent to the graph of f at $x = 1$ and the x axis
 - (b) Find the acute angle between the line $y = -3x + 13$ and the tangent to f at $x = \frac{1}{2}$

2 Differentiation

Let $f(x) = \frac{p}{x+1} + \frac{4}{11-x}, x \in [0, 10]$

1. Find values of p such that the absolute maximum occurs at $x = 10$
2. Find values of p such that the absolute minimum occurs at $x = 10$